Dimension of Development:

HEALTH AWARENESS

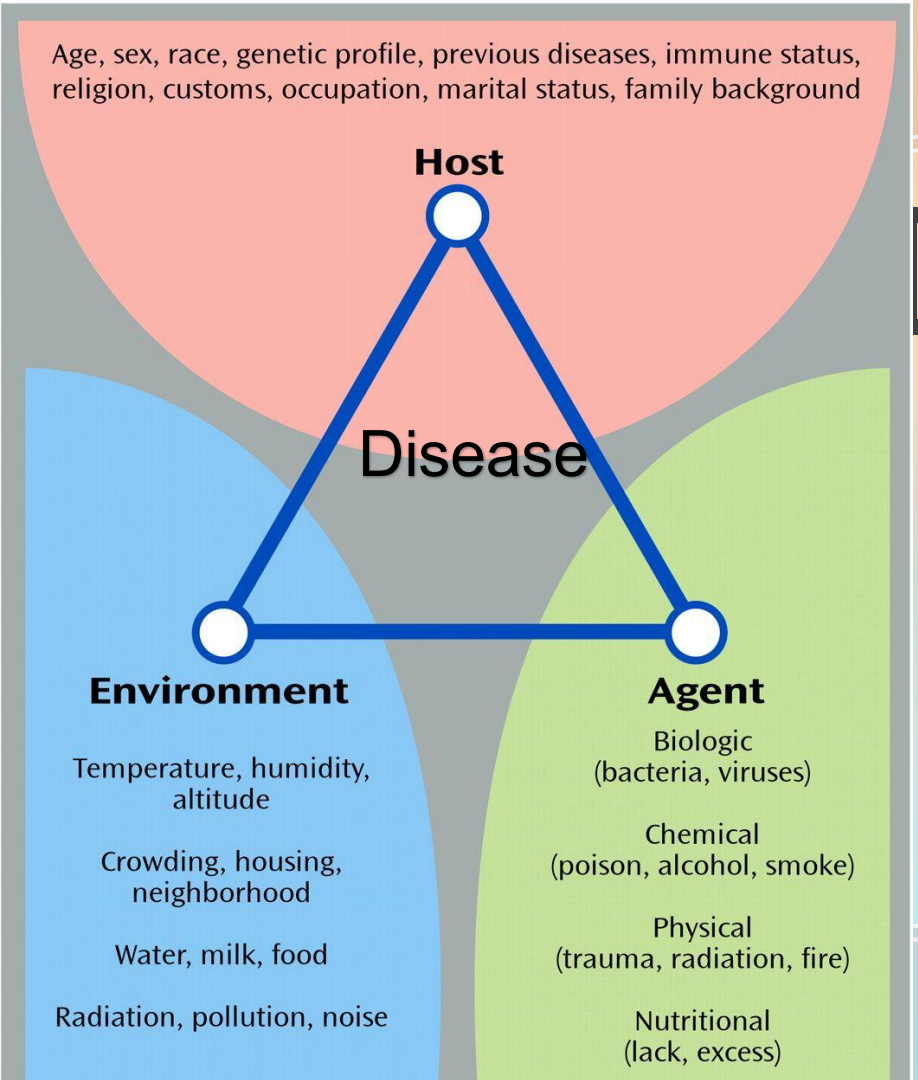
Learning Objectives:

1. Name some common communicable and non-communicable diseases.
2. Identify the causes and symptoms of these diseases.
3. Enumerate and follow the preventive measures for these diseases.



Overview

This chapter deals with some common communicable a nd non-communicable diseases and their causes, sympt oms, and prevention.



The **disease triangle** is a conceptual **model** that shows the interactions between the environment, the host and an infectious (or abiotic) agent.

This **model** can be used to predict epidemiological outcomes in plant health and public health, both in local and global communities.

Communicable Disease

- **Diseases that are spread from one person to the other.**
- **Diseases that are "catching"**
- **Diseases that are caused by germs or pathogens.**

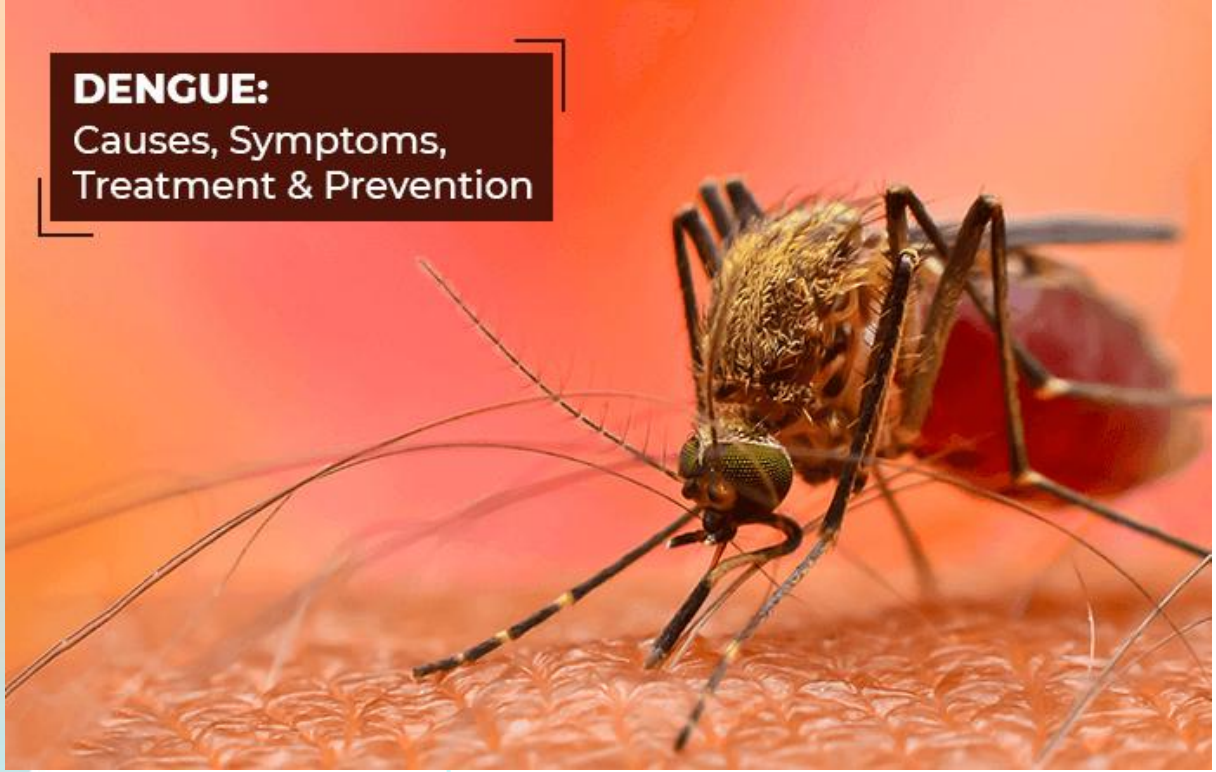


Examples of pathogens include: viruses, bacteria, p parasitic worms and fungi.



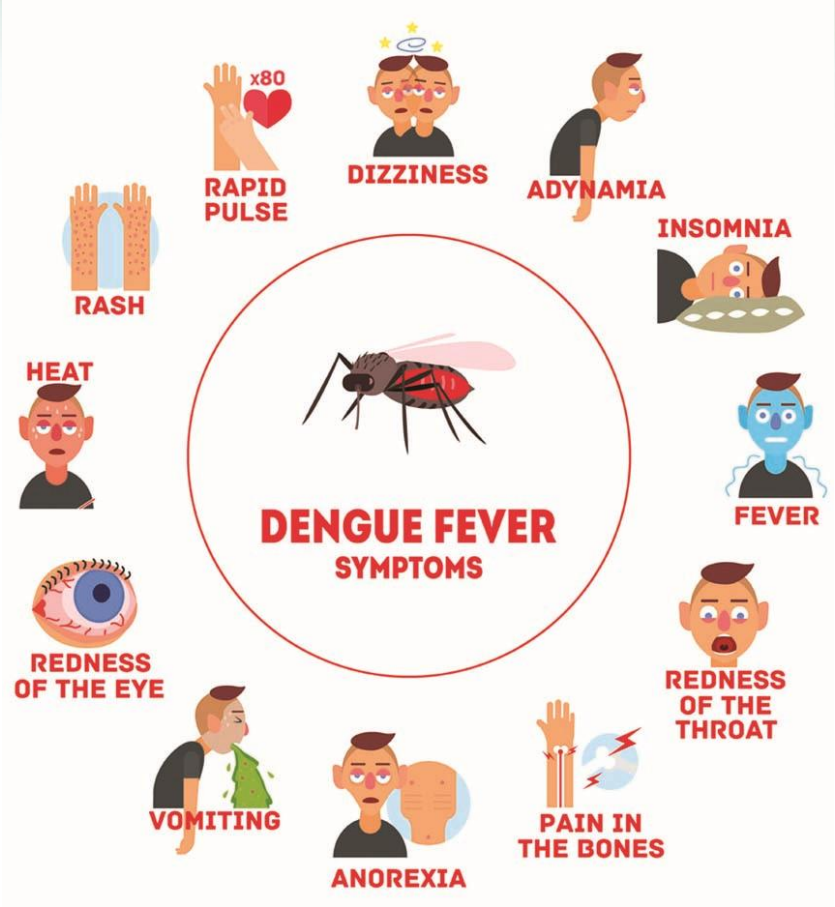
Communicable disease

Name of Disease	Mode of Spread
1. Cholera	Food and Water
2. Typhoid	Food and Water
3. Hepatitis(jaundice)	Food and Water
4. Influenza (Flu)	Air
5. Tuberculosis (T.B)	Air
6. Malaria	Mosquito
7. Tetanus	Bite Wound exposed to dust or rusted item



Dengue Fever – a febrile disease caused by one of four closely related virus serotypes of the genus ***Flavivirus***, family ***Flaviviridae.***

SIGNS AND SYMPTOMS OF DENGUE





PREVENTION

Protect yourself from mosquito bites



Daytime is most dangerous
Mosquitoes that spread chikungunya, dengue, and Zika are aggressive daytime biters. They can also bite at night.



Use insect repellent
It works!
Look for the following active ingredients:
• DEET • PICARIDIN • IR3535



Wear protective clothes
Wear long-sleeved shirts and long pants and use insect repellent. For extra protection, treat clothing with permethrin.



Mosquito-proof your home
Use screens on windows and doors. Use air conditioning when available. Keep mosquitoes from laying eggs in and near standing water.



MALARIA

is a vector-borne infectious disease that is widespread in tropical and subtropical regions. It is a serious and sometimes fatal disease caused by a parasite that commonly infects a certain type of mosquito which feeds on humans.

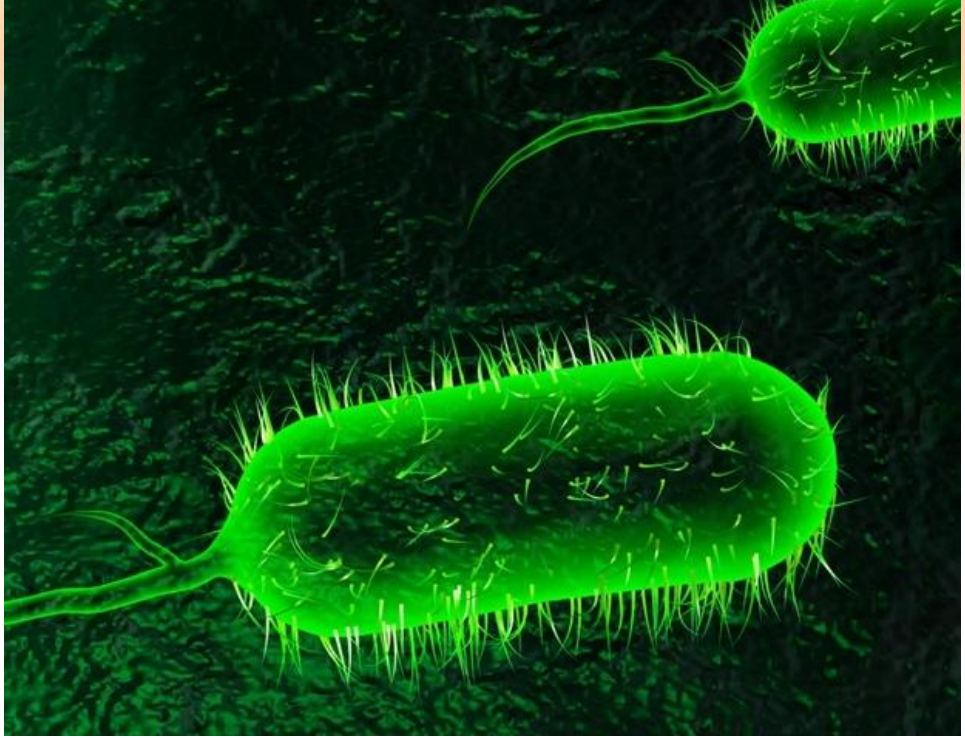
MALARIA its:

- Cause:
  - Caused by a parasite - *Plasmodium falciparum* is most serious. *P. vivax, P. ovale* and *P. malariae* are others.
  - Spread by a vector (female *Anopheles* mosquito)





- Symptoms:
  - flu-like illness, diarrhoea, vomiting, joint pains, fever.
  - Symptoms begin from a week up to a year after infection.
  - If untreated can lead to complications and death
- Treatment:
  - prescription drugs - treatment dependent on type of malaria, age of patient, and severity of illness.
  - Prevention – anti malarial drugs, avoid being bitten.



CHOLERA

An acute intestinal infection caused by ingestion of food or water contaminated with th e bacterium Vibrio cholera. It has a short incubation period lasting from less than one day to five days.

SIGNS AND SYMPTOMS



PREVENTION

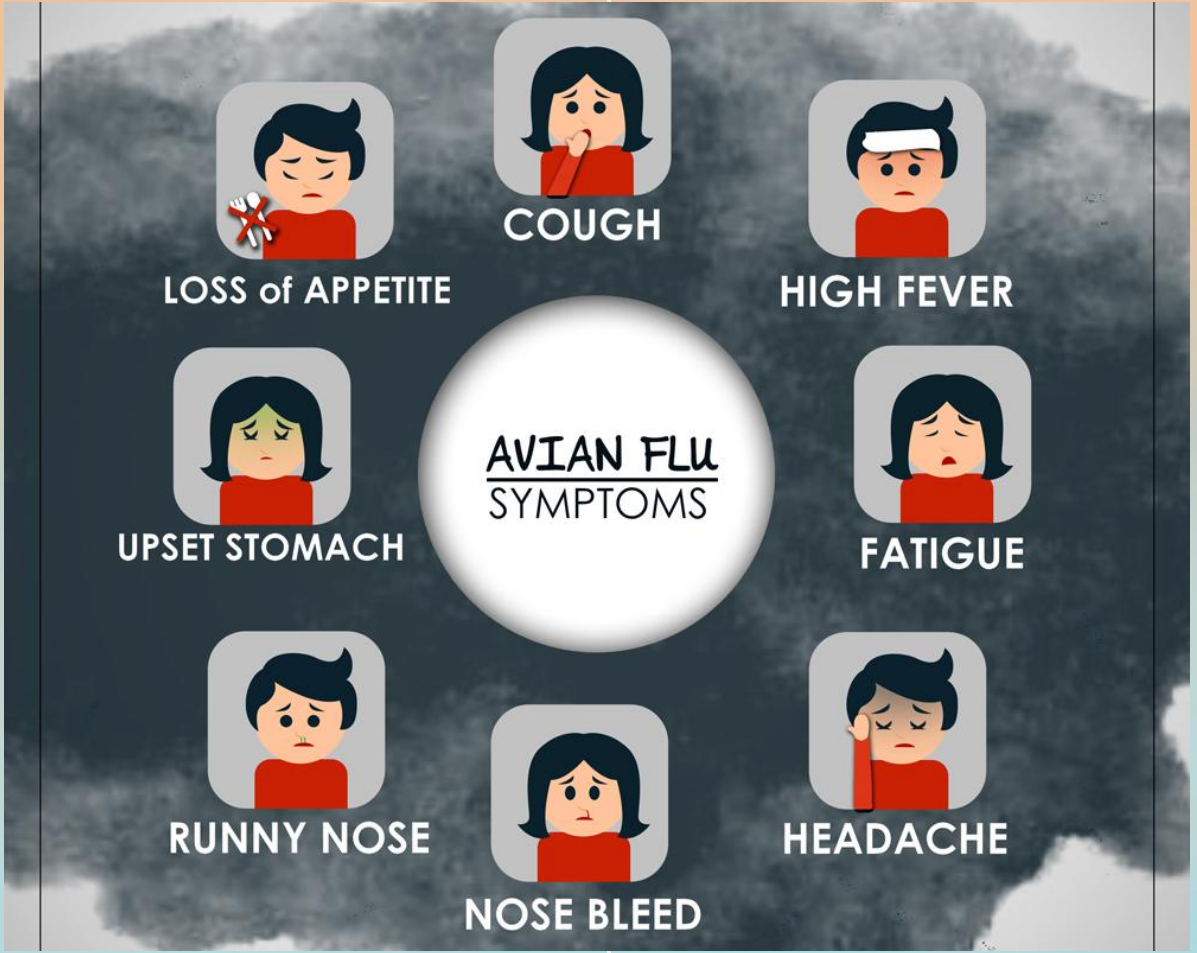

Drink safe water that is boiled or chlorinated
Wash your hands with soap and water after visiting the toilet and before handling food
Wash fruits and vegetables properly with safe water before eating and for preparing food
Cook food properly. Buy and eat only hot food



AVIAN INFLUENZA (BIRD FLU)

An influenza is an infection caused by avian influenza A viruses found chiefly in birds, but infections can occur in humans .

SIGNS and SYMPTOMS



Prevention of avian influenza

To prevent avian influenza, we should avoid touching poultry, birds and their droppings. Cook poultry and eggs thoroughly before eating, and wash hands frequently with liquid soap and water.

When going abroad, avoid visiting wet markets, live poultry markets and farms in affected areas.

If you feel unwell after the trip, wear a surgical mask. Seek medical advice promptly and reveal your travel details.

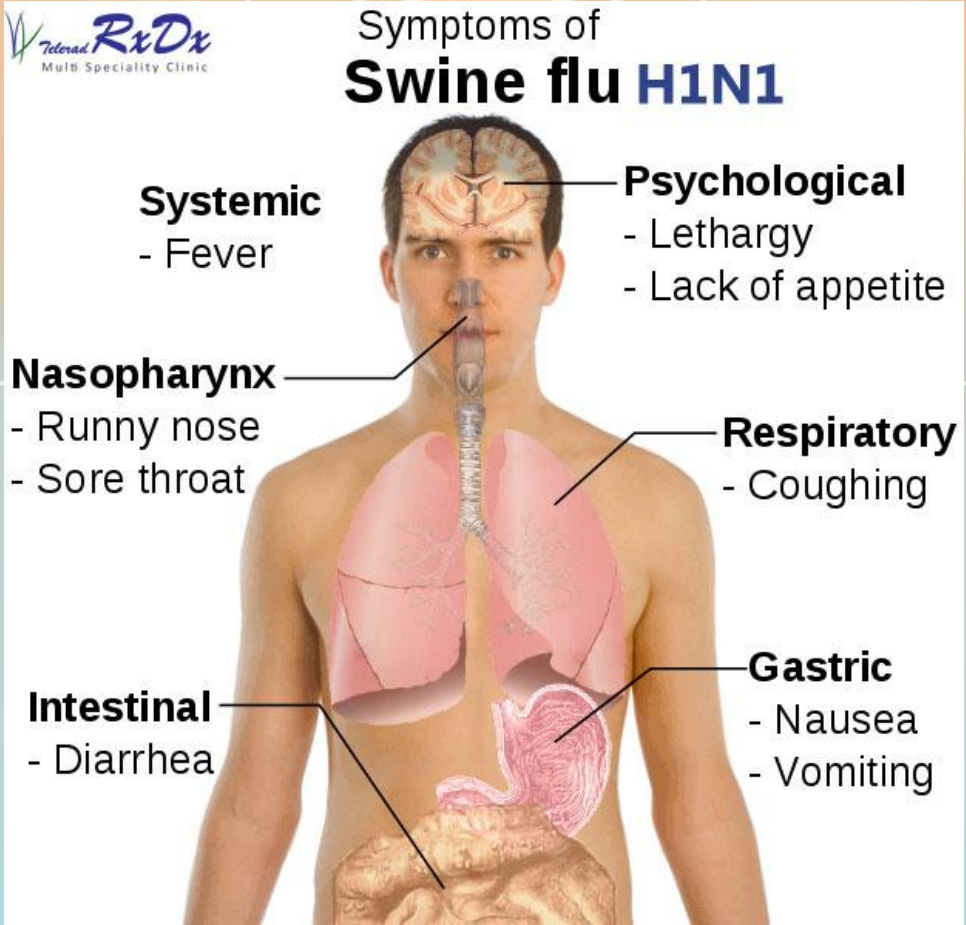
Visit chp.gov.hk to know more.



H1N1

Is a flu virus that is also known as the swine flu. Scientist call the H1N1 virus a **‘’ quadruple reassortant”** virus because it contains two genes found in flu viruses which circulate among pigs and two genes found in flu viruse s which circulates among birds.

SIGNS AND SYMPTOMS



FLU PREVENTION TIPS

GET VACCINATED Influenza is the only respiratory virus preventable by vaccination.

WASH YOUR HANDS Wash them often with soap and water.

STAY HOME Stay home for a full 24 hours after your fever goes away.

KEEP SANITIZER CLOSE Use it often and tell children and those around you to do the same.



EXERCISE It boosts your immune system and speeds recovery from illness.

DON'T TOUCH YOUR FACE This is the easiest way for germs to get into your body.

EAT RIGHT Eating healthy, balanced meals can strengthen your immune system.

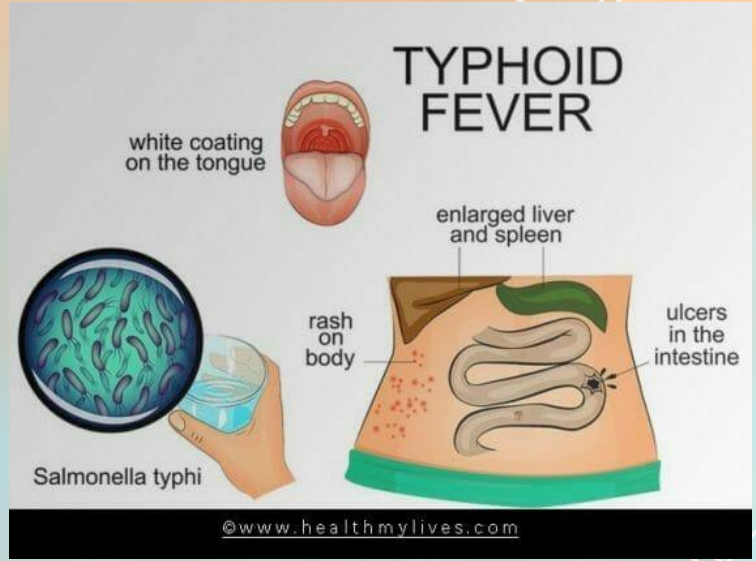
SMILE Studies show that smiling can help boost your immune system.



Also known as typhoid, an acute illness most often caused by ***Salmonella typhi*** bacteria or the ***Salmonella paratyphi***, a related bacterium that usually leads to a less severe infection.

The bacteria are deposited in water or food by a human carrier and are then spread to other people in the area.

SIGNS AND SYMPTOMS



SYMPTOMS OF TYPHOID



Malaise or weakness	Loss of appetite	Hallucinations and/or delirium
Abdominal pain	Ulcers	Watery stools/Diarrhea
Constipation	Severe headaches and irritation	High fever
Encephalitis	Dehydration	Skin rashes

PREVENTION:

- Improved personal hygiene and hand washing are important preventing measures.
- Improved sanitary habits.
- Protective health measures like provision of clean water , adequate sewage disposal and control of flies are also important.
- Clean and wash foods thoroughly.
- Adequate temperatures for cooking.
- Eggs should be thoroughly cooked and never eaten raw.
- Avoid preserving food at warm temperature and reheating food.



HIV/AIDS

•**HIV (Human Immunodeficiency Virus)**
is a virus that attacks the body's immune system, specifically the **CD4 (T-helper) cells**, which help fight infections.

•**AIDS (Acquired Immunodeficiency Syndrome)**
is the **most advanced stage** of HIV infection, when the immune system becomes severely damaged and the person is vulnerable to opportunistic infections and certain cancers.

CAUSES

HIV is transmitted through **specific body fluids** of an infecte d person:

1. **Unprotected sexual contact** (vaginal, anal, or oral) with a n infected person.
2. **Sharing needles or syringes**, especially among drug user s.
3. **Blood transfusions** with infected blood (rare today due to screening).
4. **Mother-to-child transmission** during pregnancy, childbirt h, or breastfeeding.
5. **Occupational exposure**, such as needlestick injuries in h ealthcare settings.

SYMPTOMS

1. Early Stage (Acute HIV Infection) – 2– 4 weeks after exposure

- Fever
- Fatigue
- Swollen lymph nodes
- Sore throat
- Rash
- Muscle and joint pain

(These symptoms resemble the flu and may go unnoticed.)

COMMON SYMPTOMS OF HIV



Flu-like symptoms

Flu-like symptoms: In the early stages of HIV, some people may experience symptoms similar to the flu, such as fever, fatigue, headache, muscle aches, and sore throat.

Night sweats

Night sweats: Drenching sweats during the night may occur.

Mouth sores

Mouth sores: Painful sores or ulcers in the mouth may develop.

Rash

Rash: A rash may occur on the trunk or face, and can be accompanied by itching.

Swollen lymph nodes

Swollen lymph nodes: Swelling of the lymph nodes in the neck, armpits, or groin may occur, which can persist for several weeks.

Yeast infections

Yeast infections: Yeast infections, such as thrush, may occur in the mouth, throat, or vagina.


Secondary Syphilis




EARLY HIV SYMPTOMS
Skin rash
Swollen lymph nodes
Mouth Sores
Mouth Sores

2. Clinical Latency Stage (Chronic HIV Infection)

- Often **no symptoms** for several years.
- Virus remains active but reproduces at low levels.
- Without treatment, this stage can last 5–10 years.

3. AIDS Stage

When the immune system is severely damaged:

- Rapid weight loss
- Persistent fever or night sweats
- Prolonged fatigue
- Chronic diarrhea
- Skin lesions or rashes
- Recurrent infections (e.g., tuberculosis, pneumonia)
- Memory loss or neurological problems

Effects of HIV/AIDS

1. **Immune System Failure** – HIV weakens the body's immune s ystem, making a person more vulnerable to infections and certain cancers.
2. **Neurological Problems** – HIV can affect the brain and nerves, causing memory problems, difficulty thinking, or nerve damage.
3. **Social and Psychological Impacts** – People living with HIV may experience stigma, discrimination, stress, or depression.
4. **Economic Effects** – The cost of medical care and difficulty wo rking due to illness may affect a person's income and finances.

HIV attacks the immune system, but with proper treatment, people living with HIV can live long and healthy lives.

Treatment and Management

There is **no cure** for HIV, but it can be **controlled** with proper treatment.

Antiretroviral Therapy (ART)

•The main treatment for HIV.

•Involves a combination of medicines that:

- Suppress the virus,
- Restore and protect the immune system,
- Prevent progression to AIDS,
- Reduce transmission risk.

Other Measures

- **Regular medical check-ups** and viral load testing.
- **Healthy lifestyle** — good nutrition, exercise, no smoking or alcohol abuse.
- **Psychological support** and **counseling**.
- **Preventive medications**:
  - **PrEP (Pre-Exposure Prophylaxis)** – for people at high risk of HIV.
  - **PEP (Post-Exposure Prophylaxis)** – emergency treatment within 72 hours after potential exposure.

Prevention

- Use **condoms** consistently and correctly.
- Get **regular HIV testing**.
- Avoid sharing needles or sharp instruments.
- Ensure **safe blood transfusions**.
- Pregnant women with HIV should take ART to prevent **mother-to-child transmission**.

WHAT IS YOUR FAVORITE FOOD?




PLEASE PUT WASTE IN A BIN
PLEASE PUT WASTE IN A BIN
SWEET 'N SOUR SAUCE
SWEET 'N SOUR SAUCE










GUIDE TO THE
PHILIPPINES



Non-communicable - or chronic - **diseases** are **diseases** of long duration and generally slow progression.

The 4 major categories of non-communicable diseases (NCDs)



Cardiovascular Diseases



Diabetes



Chronic Respiratory Diseases



Cancer

Hypertension



Hypertension is another name for high blood pressure. It can lead to severe health complications and increase the risk of heart disease, stroke, and sometimes death. Blood pressure is the force that a person's blood exerts against the walls of their blood vessels.

Blood pressure is recorded with **2 numbers:**

1. Systolic Pressure (Top Number) – This is the pressure in your blood vessel s when your heart contracts and pumps blood to the body.

Example: In a blood pressure reading of 120/80, 120 is the systolic pressure.

2. Diastolic Pressure (Bottom Number) – This is the pressure in your blood vess els when your heart relaxes between beats.

Example: In a blood pressure reading of 120/80, 80 is the diastolic pressure.

The top number is when the heart pumps, while the bottom number is when the heart rests between beats.


146
SYS
mmHg
88
DIA
mmHg
79
Pulse
/min
microlife
SYS
DIA
180♦
110♦
160-180
100-110
140-160
90-100
130-140
85 - 90
120-130
80 - 85
♦120
♦80

NORMAL BLOOD PRESSURE

Blood pressure reading 120/80 or lower is considered normal.

HIGH BLOOD PRESSURE

Blood pressure reading 140/90 or higher is considered high. If one or both numbers are usually high, you have high blood pressure or ***hypertension.***

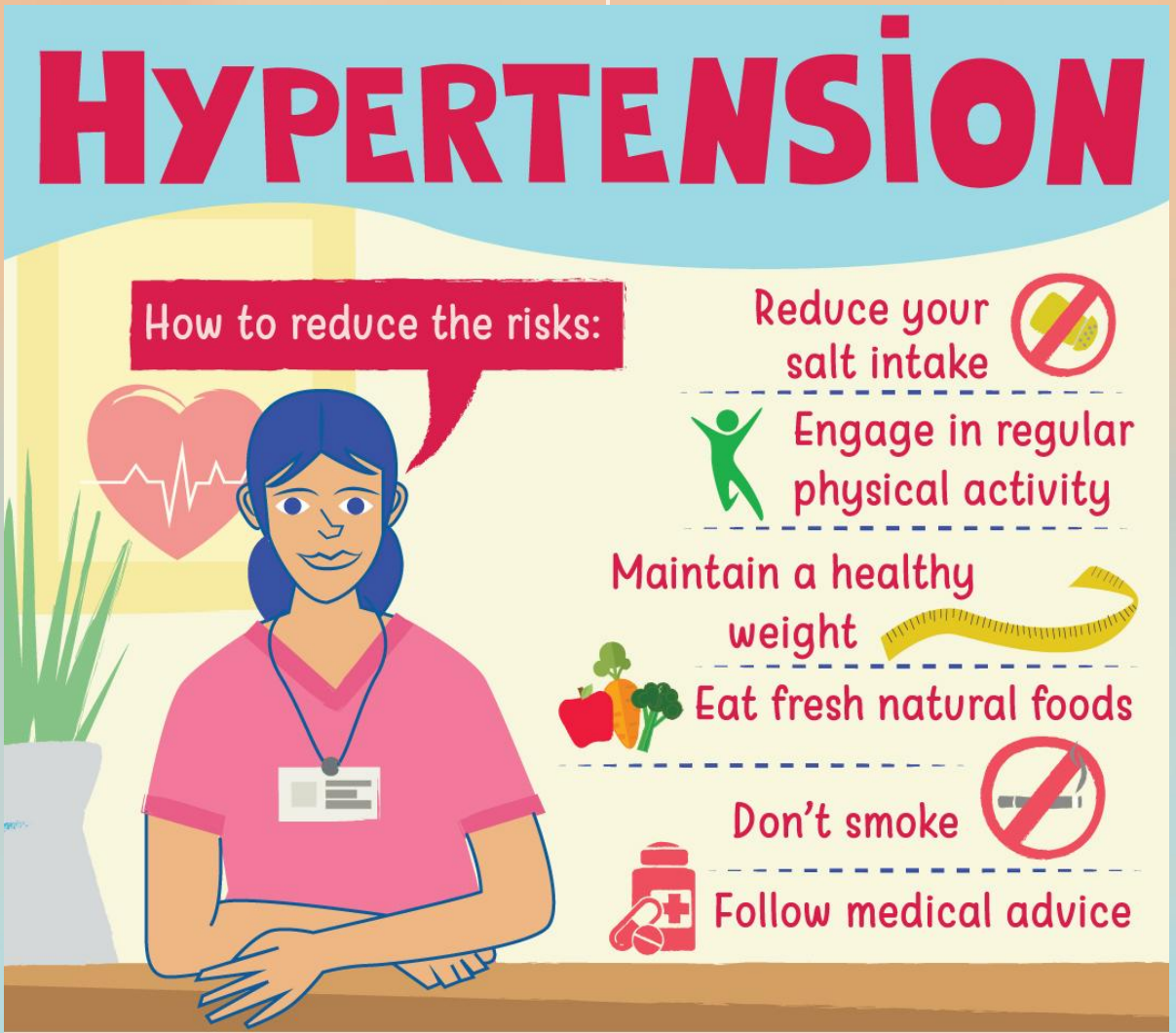
LOW BLOOD PRESSURE

Blood pressure reading 110/60 is too low is considered as ***hypotension***.

BLOOD PRESSURE CATEGORIES

Blood pressure category	Systolic mm Hg (upper number)		Diastolic mm hG (lower number)
Normal	less than 120	and	less than 80
Elevated	120-129	and	less than 80
High blood pressure (Hypertension) Stage 1	130-139	or	80-89
High blood pressure (Hypertension) Stage 2	140 or higher	or	90 or higher
Hypertensive Crisis (consult your doctor immediately)	Higher than 180	and/ or	Higher than 120

HYPERTENSION



Symptoms of High Blood Pressure (Hypertension)



Severe Headache



Vision Problems



Difficulty breathing



Fatigue



Chest Pain



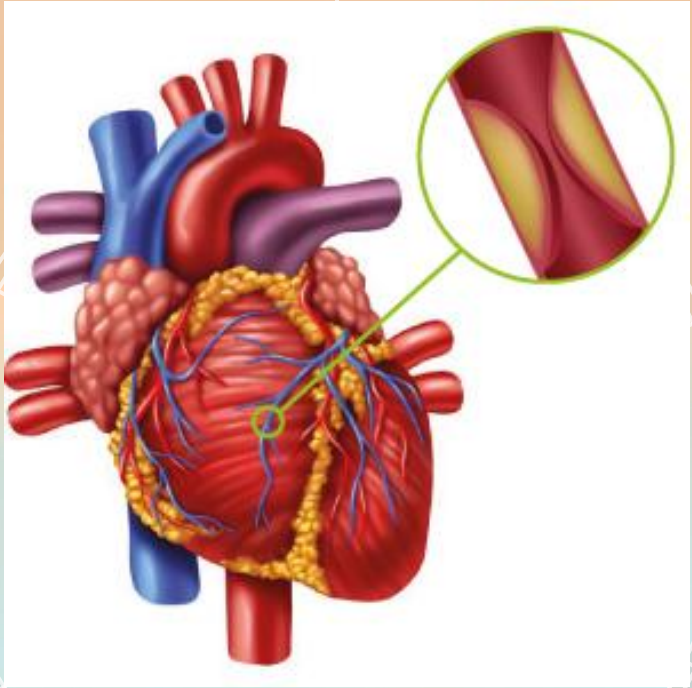
Pounding in your Chest, Neck, or Ears



Blood in the Urine



Irregular Heartbeat



Cardiovascular disease (CVD) is a general term for conditions affecting the **heart** or blood vessels. It's usually associated with a build-up of fatty deposits inside the arteries (atherosclerosis) and an increased risk of blood clots.

CAUSES OF CVD

The exact cause of CVD isn't clear, but there are lots of things that can increase your risk of getting it. These are called "risk factors".





A **heart attack** occurs when the flow of blood to the **heart** is blocked. The blockage is most often a buildup of fat, cholesterol and other substances, which form a plaque in the arteries that feed the **heart** (coronary arteries). Sometimes, a plaque can rupture and form a clot that blocks blood flow.



Stroke



A **stroke** is a sudden interruption in the blood supply of the brain. Most **strokes** are caused by an abrupt blockage of arteries leading to the brain (ischemic **stroke**).

Other **strokes** are caused by bleeding into brain tissue when a blood vessel bursts (hemorrhagic **stroke**)


10
symptoms of stroke
1. Confusion
2. Difficulty understanding
3. Dizziness
4. Loss of balance
5. Numbness
6. Severe headache
7. Trouble speaking
8. Trouble walking
9. Vision changes
10. Weakness



Diabetes is a disease in which your blood glucose, or blood sugar, levels are too high. Glucose comes from the foods you eat. Insulin is a hormone that helps the glucose get into your cells to give them energy. With type 1 **diabetes**, your body does not make insulin.

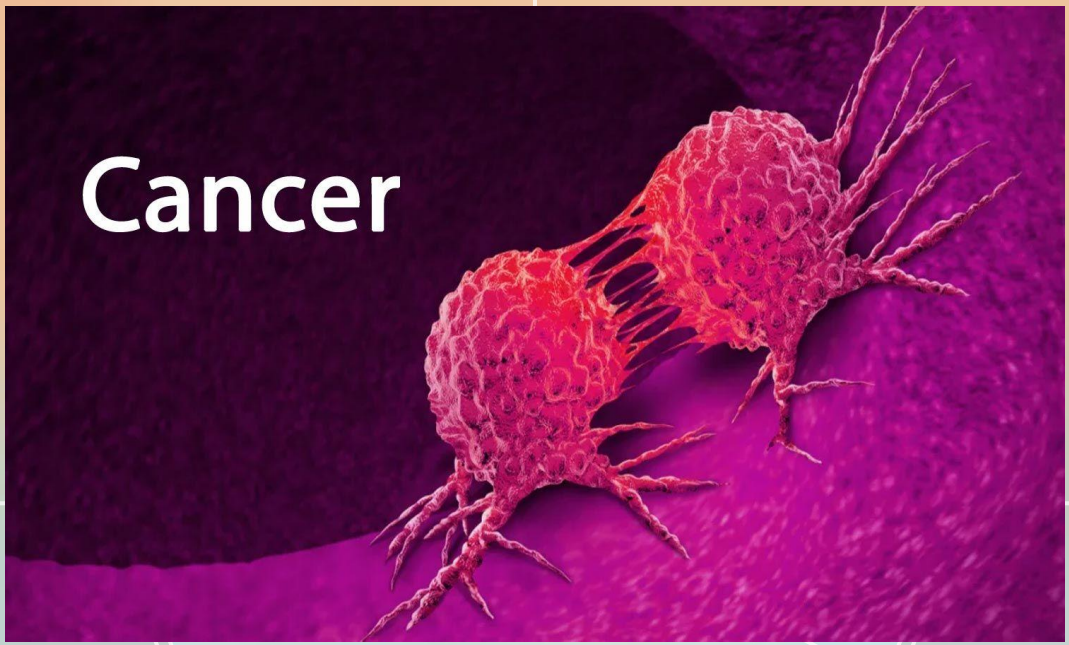
Types of diabetes

There are three main types of diabetes – **type 1, type 2 and gestational**.

Type 1 diabetes can develop at any age, but occurs most frequently in children and adolescents. When you have type 1 diabetes, your body produces very little or no insulin, which means that you need daily insulin injections to maintain blood glucose levels under control.

Type 2 diabetes is more common in adults and accounts for around 90% of all diabetes cases. When you have type 2 diabetes, your body does not make good use of the insulin that it produces. The cornerstone of type 2 diabetes treatment is healthy lifestyle, including increased physical activity and healthy diet. However, over time most people with type 2 diabetes will require oral drugs and/or insulin to keep their blood glucose levels under control.

Gestational diabetes (GDM) is a type of diabetes that consists of high blood glucose during pregnancy and is associated with complications to both mother and child. GDM usually disappears after pregnancy but women affected and their children are at increased risk of developing type 2 diabetes later in life.



Cancer is a group of diseases involving abnormal cell growth with the potential to invade or spread to other parts of the body. These contrast with benign tumors, which do not spread.

CLASSIFICATION OF CANCER

Carcinomas, Sarcomas, Leukemia, Lymphomas, Adenomas

Carcinomas. A carcinoma begins in the skin or the tissue that covers the surface of internal organs and glands. Carcinomas usually form solid tumors. They are the most common type of cancer.

Examples of carcinomas include **prostate cancer**, **breast cancer**, **lung cancer**, and **colorectal cancer**.



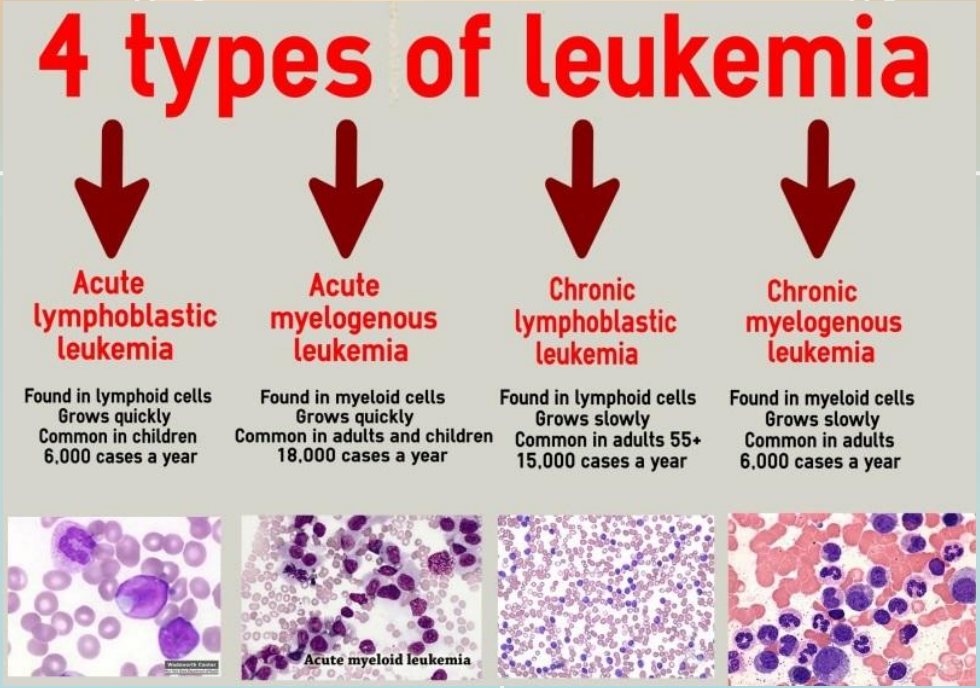
Sarcomas. A **sarcoma** begins in the tissues that support and connect the body. A sarcoma can develop in fat, muscles, nerves, tendons, joints, blood vessels, lymph vessels, cartilage, or bone.





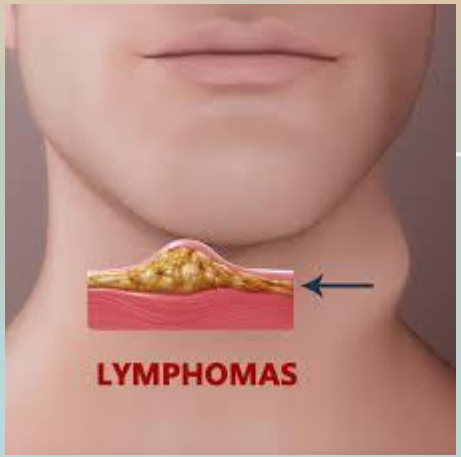
•**Leukemia.** Leukemia is a cancer of the blood. Leukemia begins when healthy blood cells change and grow uncontrollably.

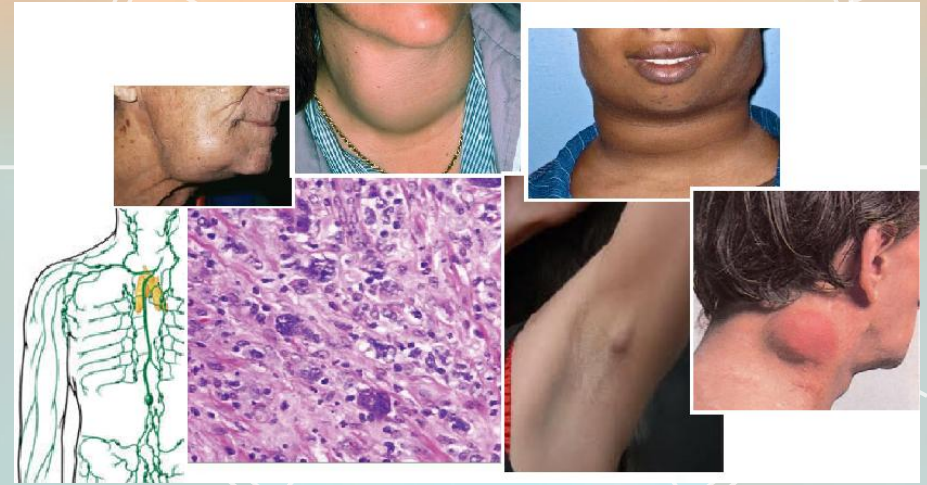
The 4 main types of leukemia are **acute lymphocytic leukemia**, **chronic lymphocytic leukemia**, **acute myeloid leukemia**, and **chronic myeloid leukemia**.

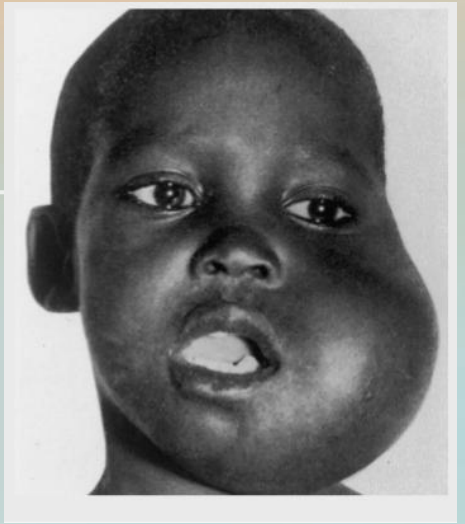


•***Lymphomas.*** Lymphoma is a cancer that begins in the lymphatic system. The lymphatic system is a network of vessels and glands that help fight infection.

There are 2 main types of lymphomas: **Hodgkin lymphoma** and **non-Hodgkin lymphoma**.





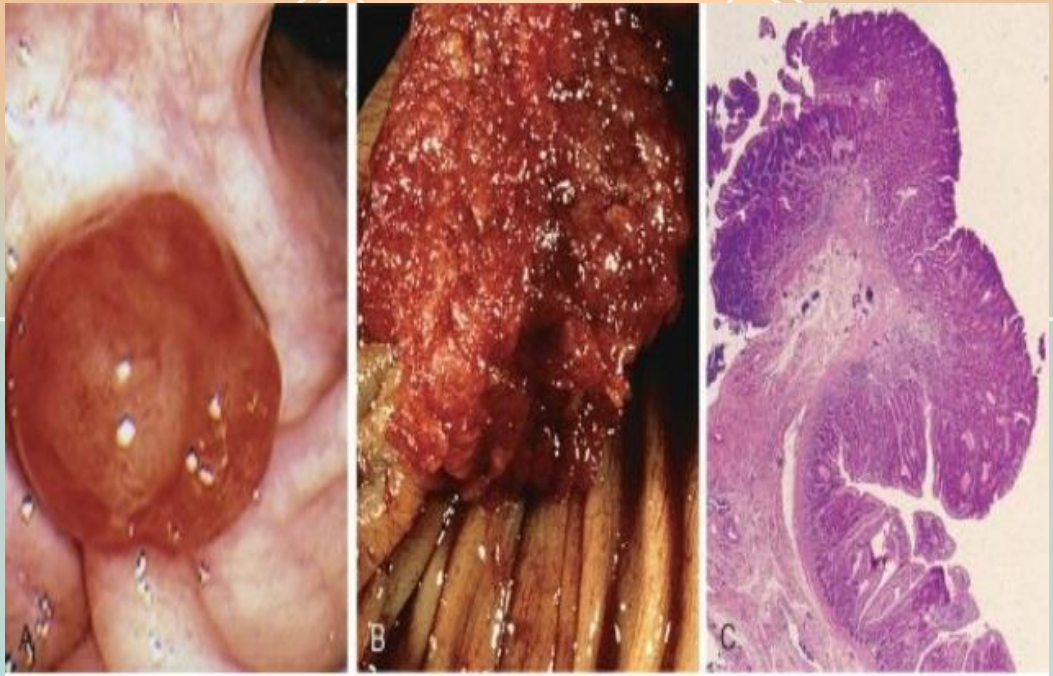


1. Hodgkin Lymphoma
•Has a specific type of abnormal cell called the Reed-Sternberg cell.
•Often spreads in an orderly way from one group of lymph nodes to another.

2. Non-Hodgkin Lymphoma
•Usually does not have Reed-Sternberg cells.
•Can spread in different patterns and may develop from B cells or T cells.

The main difference is the presence of Reed-Sternberg cells in Hodgkin lymphoma. Both types require medical diagnosis and treatment.

•***Adenomas*** are cancers that arise in the thyroid, pituitary gland, adrenal gland, and other glandular tissues.



Colonic adenomas. A, Pedunculated adenoma .B, Adenoma with a velvety surface. C, Low-magnification photomicrograph of a pedunculated tubular adenoma.

Definitions

Benign Tumors	**Malignant** Tumors
• Small • Slow-growing • Non-invasive • Well-differentiated • Stay localized • Stay where they are. • Can't invade or metastasize.	• Large • Fast-growing • Invasive • Poorly-differentiated • Metastasize • Infiltrate, invade, destroy surrounding tissue. • Then metastasize to other parts of body.

Early Cancer Warning Sign

One symptom you shouldn't ignore:

unexplained weight loss



